

BEC303 Network Analysis — Module 5 Quiz

Two-Port Networks

Field	Value
Subject code	BEC303
Module	5 — Two-Port Networks
Duration	45 minutes
Maximum marks	50 (Part A 20 + Part B 30)
Attempts allowed	1
Passing marks	20
Textbook	W. H. Hayt, J. E. Kemmerly and S. M. Durbin, Engineering Circuit Analysis, 8th ed., 2012

Instructions

- Part A carries 20 one-mark multiple choice questions. Exactly one option is correct.
- Part B carries 13 short questions totalling 30 marks.
- State the termination condition used for every parameter you compute.
- Give units for every parameter; mixed-unit sets must be labelled explicitly.
- Non-programmable scientific calculators are permitted.

Course outcome and Bloom coverage

Band	Questions	Share	Target
L1–L2 Remember / Understand	6	18%	~30%
L3–L4 Apply / Analyse	22	67%	~45%
L5–L6 Evaluate / Create	5	15%	~25%

Course outcome spread: CO5: 33. Total questions: 33.

Part A — Multiple choice (20 × 1 mark = 20 marks)

A1. The port condition requires that [C05 | L2 | 1 mark | medium | Ch.17, pp.687-692]

- (A) the current entering one terminal equals the current leaving the other
- (B) both terminals be at the same potential
- (C) the port be terminated in a matched load
- (D) the network be reciprocal

A2. How many independent parameters characterise a linear two-port at a given frequency? [C05 | L2 | 1 mark | easy | Ch.17, pp.687-692]

- (A) Four
- (B) Two

- (C) Six
- (D) It depends on the number of internal elements

A3. The y parameters are measured with [C05 | L2 | 1 mark | easy | Ch.17, pp.692-699]

- (A) one port short-circuited
- (B) one port open-circuited
- (C) both ports matched
- (D) the network unterminated

A4. z_{11} is defined as [C05 | L2 | 1 mark | medium | Ch.17, pp.708-713]

- (A) V_1/I_1 with port 2 open-circuited
- (B) V_1/I_1 with port 2 short-circuited
- (C) V_1/I_1 with port 2 matched
- (D) the input impedance under any load

A5. The y parameters map naturally onto which equivalent circuit? [C05 | L3 | 1 mark | medium | Ch.17, pp.699-708]

- (A) The π network
- (B) The T network
- (C) The ladder network
- (D) The bridged-T network

A6. For $z = [[30, 20], [20, 30]] \Omega$, the determinant Δz is [C05 | L2 | 1 mark | easy | Ch.17, pp.692-713]

- (A) 500
- (B) 900
- (C) 400
- (D) 1300

A7. To convert a z matrix to a y matrix you must [C05 | L3 | 1 mark | medium | Ch.17, pp.692-713]

- (A) invert the matrix
- (B) reciprocate each element
- (C) transpose the matrix
- (D) divide by the trace

A8. The h parameters have mixed units because [C05 | L3 | 1 mark | medium | Ch.17, pp.713-716]

- (A) they relate mixed variables — V_1 and I_2 to I_1 and V_2
- (B) they were defined before SI units existed
- (C) they apply only to nonlinear devices
- (D) they are measured at mixed frequencies

A9. In transistor datasheets, h_{21} corresponds to [C05 | L2 | 1 mark | medium | Ch.17, pp.713-716]

- (A) h_{fe} , the forward current gain
- (B) h_{ie} , the input impedance
- (C) the output conductance
- (D) the reverse voltage ratio

A10. A lone series impedance between two ports has [C05 | L4 | 1 mark | hard | Ch.17, pp.692-716]

- (A) no z representation, because opening a port leaves no current path
- (B) no y representation
- (C) neither a z nor an ABCD representation
- (D) all six representations

A11. The ABCD matrix of a single series impedance Z is [C05 | L3 | 1 mark | medium | Ch.17, pp.716-720]

- (A) $\begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix}$
- (B) $\begin{bmatrix} 1 & 0 \\ Z & 1 \end{bmatrix}$
- (C) $\begin{bmatrix} Z & 1 \\ 1 & 0 \end{bmatrix}$
- (D) $\begin{bmatrix} 1 & 1 \\ Z & Z \end{bmatrix}$

A12. For cascaded two-ports the overall ABCD matrix is [C05 | L3 | 1 mark | medium | Ch.17, pp.716-720]

- (A) the ordered product of the individual matrices
- (B) the sum of the individual matrices
- (C) the product in any order
- (D) the inverse of the product

A13. Cascading a series $10\ \Omega$ with a shunt $0.05\ \text{S}$ gives an overall ABCD of [C05 | L4 | 1 mark | hard | Ch.17, pp.716-720]

- (A) $\begin{bmatrix} 1.5 & 10 \\ 0.05 & 1 \end{bmatrix}$
- (B) $\begin{bmatrix} 1 & 10 \\ 0.05 & 1.5 \end{bmatrix}$
- (C) $\begin{bmatrix} 1.5 & 10 \\ 0.05 & 1.5 \end{bmatrix}$
- (D) $\begin{bmatrix} 1 & 10 \\ 0.05 & 1 \end{bmatrix}$

A14. The reciprocity condition for ABCD parameters is [C05 | L3 | 1 mark | medium | Ch.17, pp.699-720]

- (A) $AD - BC = 1$
- (B) $A = D$
- (C) $B = C$
- (D) $A + D = 2$

A15. A network with $z = \begin{bmatrix} 30 & 20 \\ 20 & 45 \end{bmatrix}\ \Omega$ is [C05 | L4 | 1 mark | hard | Ch.17, pp.699-708]

- (A) reciprocal but not symmetric
- (B) symmetric but not reciprocal
- (C) both reciprocal and symmetric
- (D) neither

A16. The input impedance of a terminated two-port is [C05 | L3 | 1 mark | medium | Ch.17, pp.708-713]

- (A) $z_{11} - z_{12}z_{21}/(z_{22} + Z_L)$
- (B) z_{11}
- (C) $z_{11} + Z_L$
- (D) $z_{11}z_{22}/(z_{12}z_{21})$

A17. For $z = \begin{bmatrix} 30 & 20 \\ 20 & 30 \end{bmatrix}\ \Omega$ with a $50\ \Omega$ load, Z_{in} is [C05 | L4 | 1 mark | hard | Ch.17, pp.708-713]

- (A) $25\ \Omega$

- (B) $30\ \Omega$
- (C) $80\ \Omega$
- (D) $16.7\ \Omega$

A18. The output impedance of a two-port depends on [C05 | L4 | 1 mark | hard | Ch.17, pp.708-713]

- (A) the source impedance at port 1
- (B) the load impedance at port 2
- (C) neither termination
- (D) the frequency only

A19. Why do scattering parameters replace y parameters at high frequency? [C05 | L5 | 1 mark | hard | Ch.17, pp.692-713]

- (A) A true short circuit is hard to realise because of lead inductance
- (B) y parameters do not exist above a certain frequency
- (C) Scattering parameters have simpler units
- (D) y parameters cannot describe reciprocal networks

A20. Two-ports connected in parallel have which matrices that add? [C05 | L4 | 1 mark | hard | Ch.17, pp.692-720]

- (A) y
- (B) z
- (C) h
- (D) ABCD

Part B — Descriptive (13 questions, 30 marks)

B1. Define the z parameters, state the termination used for each, and give the corresponding equivalent circuit. [C05 | L3 | 3 marks | Ch.17, pp.708-713]

B2. Find the z parameters of a T network with $10\ \Omega$ series arms and a $20\ \Omega$ shunt arm, and verify reciprocity and symmetry. [C05 | L3 | 3 marks | Ch.17, pp.708-713]

B3. Convert $z = \begin{bmatrix} 30 & 20 \\ 20 & 30 \end{bmatrix}\ \Omega$ to y parameters and explain why elementwise reciprocation is wrong. [C05 | L4 | 3 marks | Ch.17, pp.692-713]

B4. Explain why six parameter sets exist, giving one reason of each of the three distinct kinds. [C05 | L5 | 3 marks | Ch.17, pp.692-720]

B5. State the ABCD defining equations, explain the minus signs, and give the matrix for a series impedance Z. [C05 | L4 | 2 marks | Ch.17, pp.716-720]

B6. Cascade a series $10\ \Omega$ with a shunt $0.05\ \text{S}$ and verify the result. [C05 | L4 | 2 marks | Ch.17, pp.716-720]

B7. Distinguish reciprocity from symmetry, giving the condition for each in z and in ABCD parameters. [C05 | L4 | 2 marks | Ch.17, pp.699-720]

B8. Derive the π equivalent from $y = \begin{bmatrix} 0.15 & -0.1 \\ -0.1 & 0.15 \end{bmatrix}\ \text{S}$. [C05 | L4 | 2 marks | Ch.17, pp.699-708]

B9. Explain the h parameters and why their mixed units are unavoidable. [C05 | L3 | 2 marks | Ch.17, pp.713-716]

B10. Compute Z_{in} for $z = [[30, 20], [20, 30]] \Omega$ with a 50Ω load, and verify with both limiting cases. [C05 | L4 | 2 marks | Ch.17, pp.708-713]

B11. Compute Z_{out} for $z = [[30, 20], [20, 30]] \Omega$ driven from a 10Ω source, and state the matched load. [C05 | L5 | 2 marks | Ch.17, pp.708-713]

B12. Explain why a short circuit becomes unusable as a measurement termination at high frequency, with a numerical estimate. [C05 | L5 | 2 marks | Ch.17, pp.692-713]

B13. For $z = [[30, 20], [20, 30]] \Omega$ with $Z_S = 10 \Omega$ and $Z_L = 50 \Omega$, compute the voltage gain and comment on the output match. [C05 | L5 | 2 marks | Ch.17, pp.708-720]

Answer key — Part A

Q	Answer	CO	Bloom	Reference
A1	A	CO5	L2	Ch.17, pp.687-692
A2	A	CO5	L2	Ch.17, pp.687-692
A3	A	CO5	L2	Ch.17, pp.692-699
A4	A	CO5	L2	Ch.17, pp.708-713
A5	A	CO5	L3	Ch.17, pp.699-708
A6	A	CO5	L2	Ch.17, pp.692-713
A7	A	CO5	L3	Ch.17, pp.692-713
A8	A	CO5	L3	Ch.17, pp.713-716
A9	A	CO5	L2	Ch.17, pp.713-716
A10	A	CO5	L4	Ch.17, pp.692-716
A11	A	CO5	L3	Ch.17, pp.716-720
A12	A	CO5	L3	Ch.17, pp.716-720
A13	A	CO5	L4	Ch.17, pp.716-720
A14	A	CO5	L3	Ch.17, pp.699-720
A15	A	CO5	L4	Ch.17, pp.699-708
A16	A	CO5	L3	Ch.17, pp.708-713
A17	A	CO5	L4	Ch.17, pp.708-713
A18	A	CO5	L4	Ch.17, pp.708-713
A19	A	CO5	L5	Ch.17, pp.692-713
A20	A	CO5	L4	Ch.17, pp.692-720

Part A — why each option is right or wrong

A1. Correct answer: (A) the current entering one terminal equals the current leaving the other

- (A) Correct — without it, no single impedance describes the terminal pair.
- (B) Wrong - equal potentials would mean zero port voltage, which is a short circuit.
- (C) Wrong - matching is a design choice, not a requirement for the pair to be a port.
- (D) Wrong - reciprocity is a separate property and is not needed for the port condition.

A2. Correct answer: (A) Four

- (A) Correct — two of the four port variables are expressed in terms of the other two, needing four coefficients.
- (B) Wrong - two coefficients would describe a one-port, not a two-port.
- (C) Wrong - six is the number of common parameter sets, not the number of parameters in each.
- (D) Wrong - the internal element count is exactly what the abstraction discards.

A3. Correct answer: (A) one port short-circuited

- (A) Correct — setting a port voltage to zero isolates a single coefficient.
- (B) Wrong - open circuits set a port current to zero, which is the z-parameter condition.
- (C) Wrong - matched terminations are the scattering-parameter condition.
- (D) Wrong - some definite termination is always needed to isolate individual coefficients.

A4. Correct answer: (A) V_1/I_1 with port 2 open-circuited

- (A) Correct — the open circuit forces $I_2 = 0$, isolating the coefficient.
- (B) Wrong - a short gives a different and generally much smaller number.
- (C) Wrong - matching is not the defining condition for z parameters.
- (D) Wrong - the terminated input impedance depends on the load and is not z_{11} .

A5. Correct answer: (A) The π network

- (A) Correct — shunt admittances add, which is what y parameters describe.
- (B) Wrong - the T network is the natural equivalent for z parameters.
- (C) Wrong - a ladder is a cascade and is naturally handled by ABCD parameters.
- (D) Wrong - the bridged-T is a specific topology and not the generic equivalent for any parameter set.

A6. Correct answer: (A) 500

- (A) Correct — $30 \times 30 - 20 \times 20 = 900 - 400 = 500$.
- (B) Wrong - 900 is the product of the diagonal alone, with the off-diagonal term not subtracted.
- (C) Wrong - 400 is the off-diagonal product alone.
- (D) Wrong - this adds the two products instead of subtracting.

A7. Correct answer: (A) invert the matrix

- (A) Correct — $y = z^{-1}$, provided the determinant is non-zero.
- (B) Wrong - elementwise reciprocation is not matrix inversion; for a typical z it is off by tens of percent.
- (C) Wrong - transposition swaps the off-diagonal terms and does nothing useful here.
- (D) Wrong - the trace plays no role in a 2x2 inversion.

A8. Correct answer: (A) they relate mixed variables — V_1 and I_2 to I_1 and V_2

- (A) Correct — h_{11} is in ohms, h_{22} in siemens, and the other two are dimensionless.
- (B) Wrong - the mixed units follow from the variable choice, not from historical accident.
- (C) Wrong - h parameters describe linear small-signal models.
- (D) Wrong - all four are measured at the same frequency.

A9. Correct answer: (A) h_{fe} , the forward current gain

- (A) Correct — h_{21} is the short-circuit forward current gain.
- (B) Wrong - h_{ie} corresponds to h_{11} , the input impedance.
- (C) Wrong - the output conductance is h_{22} .
- (D) Wrong - the reverse voltage ratio is h_{12} .

A10. Correct answer: (A) no z representation, because opening a port leaves no current path

- (A) Correct — this is the standard example of why more than one parameter set is needed.
- (B) Wrong - the y matrix exists for this network; it is z that fails.
- (C) Wrong - its ABCD matrix is the perfectly well-defined $\begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix}$.
- (D) Wrong - if all six existed there would be no reason for six sets to exist.

A11. Correct answer: (A) $\begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix}$

- (A) Correct — an open output gives $A = 1$ and $C = 0$; a shorted output gives $B = Z$ and $D = 1$.
- (B) Wrong - that is the shunt-admittance matrix with Y in the lower-left position.
- (C) Wrong - this violates the $AD - BC = 1$ reciprocity condition.
- (D) Wrong - this is dimensionally inconsistent and fails every check.

A12. Correct answer: (A) the ordered product of the individual matrices

- (A) Correct — the order must follow signal flow, since matrix multiplication does not commute.
- (B) Wrong - addition applies to y for parallel and z for series connections, not to cascades.
- (C) Wrong - reversing the order describes a physically different cascade.
- (D) Wrong - inversion corresponds to looking through the network backwards, not to cascading.

A13. Correct answer: (A) $\begin{bmatrix} 1.5 & 10 \\ 0.05 & 1 \end{bmatrix}$

- (A) Correct — the product $\begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0.05 & 1 \end{bmatrix}$ gives $A = 1.5$ and $D = 1$, and $AD - BC = 1$.
- (B) Wrong - the 1.5 appears in the A position, not the D position, for this order.
- (C) Wrong - this would give $AD - BC = 2.25 - 0.5$, which fails the reciprocity check.
- (D) Wrong - this omits the loading interaction between the two elements entirely.

A14. Correct answer: (A) $AD - BC = 1$

- (A) Correct — this is the ABCD form of the reciprocity statement.
- (B) Wrong - $A = D$ is the symmetry condition, which is a different claim.
- (C) Wrong - B and C have different units and cannot be equal.
- (D) Wrong - this has no standing as either a reciprocity or a symmetry condition.

A15. Correct answer: (A) reciprocal but not symmetric

- (A) Correct — $z_{12} = z_{21}$ gives reciprocity, but $z_{11} \neq z_{22}$ so it is not symmetric.
- (B) Wrong - the symmetry condition is the one that fails here, not the reciprocity condition.
- (C) Wrong - symmetry requires equal diagonal entries, and 30 is not 45.
- (D) Wrong - the equal off-diagonal entries do establish reciprocity.

A16. Correct answer: (A) $z_{11} - z_{12}z_{21}/(z_{22} + Z_L)$

- (A) Correct — the second term is the load reflected back through the network.
- (B) Wrong - z_{11} is the open-circuit value only and ignores the load entirely.
- (C) Wrong - the load does not simply add in series; it is reflected through the transfer terms.
- (D) Wrong - this expression has no physical basis and is dimensionally wrong.

A17. Correct answer: (A) 25 Ω

- (A) Correct — $30 - (20 \times 20)/(30 + 50) = 30 - 5 = 25 \Omega$.

- (B) Wrong - 30 ohm is z_{11} , the value only for an open-circuited output.
- (C) Wrong - this adds the load to z_{11} , which is not how the load enters.
- (D) Wrong - 16.7 ohm is the value for a short-circuited output, not a 50 ohm load.

A18. Correct answer: (A) the source impedance at port 1

- (A) Correct — $Z_{\text{out}} = z_{22} - z_{12}z_{21}/(z_{11} + Z_S)$, reflecting the source through the network.
- (B) Wrong - the load is what Z_{out} is presented to; it cannot also determine it.
- (C) Wrong - each terminated impedance depends on the termination at the far port.
- (D) Wrong - frequency matters, but so does the source impedance.

A19. Correct answer: (A) A true short circuit is hard to realise because of lead inductance

- (A) Correct — lead inductance makes the short an impedance that rises with frequency, corrupting the measurement.
- (B) Wrong - the parameters exist mathematically; it is the measurement that becomes impractical.
- (C) Wrong - scattering parameters are dimensionless, but that is not the motivating reason.
- (D) Wrong - y parameters describe reciprocal networks perfectly well.

A20. Correct answer: (A) y

- (A) Correct — parallel connection puts the same voltages across both, so the currents and therefore the admittances add.
- (B) Wrong - z matrices add for a series connection, not a parallel one.
- (C) Wrong - h matrices add for a series-parallel connection, a different arrangement again.
- (D) Wrong - ABCD matrices multiply for a cascade; they never add.

Part B — model answers and marking scheme

B1 (3 marks) — CO5, L3, Ch.17, pp.708-713

Question. Define the z parameters, state the termination used for each, and give the corresponding equivalent circuit.

Model answer. The z parameters relate both voltages to both currents: $V_1 = z_{11}I_1 + z_{12}I_2$ and $V_2 = z_{21}I_1 + z_{22}I_2$. Each is measured with one port open-circuited: $z_{11} = V_1/I_1$ and $z_{21} = V_2/I_1$ with port 2 open; $z_{22} = V_2/I_2$ and $z_{12} = V_1/I_2$ with port 1 open. The corresponding equivalent circuit is the T network, with series arms $z_{11} - z_{12}$ and $z_{22} - z_{12}$ and a shunt arm z_{12} . All four have units of ohms and are frequency-dependent.

Point required	Marks
Both defining equations correct	1
Correct open-circuit conditions for all four	1
Correct T model with its arm values	1
Total	3

B2 (3 marks) — CO5, L3, Ch.17, pp.708-713

Question. Find the z parameters of a T network with 10 Ω series arms and a 20 Ω shunt arm, and verify reciprocity and symmetry.

Model answer. Open port 2 and drive port 1: the current passes through the $10\ \Omega$ series arm and the $20\ \Omega$ shunt arm, so $z_{11} = 30\ \Omega$. The port 2 voltage is the drop across the shunt arm only, so $z_{21} = 20\ \Omega$. By the network's geometric symmetry, $z_{22} = 30\ \Omega$ and $z_{12} = 20\ \Omega$. So $z = \begin{bmatrix} 30 & 20 \\ 20 & 30 \end{bmatrix}\ \Omega$. Reciprocity holds since $z_{12} = z_{21}$, and symmetry holds since $z_{11} = z_{22}$.

Point required	Marks
Correct z_{11} and z_{21} with the open-circuit reasoning	2
Remaining parameters plus both checks	1
Total	3

B3 (3 marks) — CO5, L4, Ch.17, pp.692-713

Question. Convert $z = \begin{bmatrix} 30 & 20 \\ 20 & 30 \end{bmatrix}\ \Omega$ to y parameters and explain why elementwise reciprocation is wrong.

Model answer. The conversion is a matrix inversion. $\Delta z = 900 - 400 = 500$. Then $y_{11} = z_{22}/\Delta z = 0.06\ \text{S}$, $y_{12} = -z_{12}/\Delta z = -0.04\ \text{S}$, $y_{21} = -0.04\ \text{S}$ and $y_{22} = 0.06\ \text{S}$. Elementwise reciprocation would give $y_{11} = 1/30 = 0.0333\ \text{S}$, which is 80 % below the correct $0.06\ \text{S}$, because inversion mixes all four entries through the determinant rather than treating them independently.

Point required	Marks
Correct determinant and all four y entries	2
Correct explanation of why reciprocation fails, with the numerical comparison	1
Total	3

B4 (3 marks) — CO5, L5, Ch.17, pp.692-720

Question. Explain why six parameter sets exist, giving one reason of each of the three distinct kinds.

Model answer. First, existence: some networks have a singular matrix in one representation, so a lone series impedance has no z parameters because opening a port leaves no current path. Second, measurement practicality: a true short circuit is hard to realise at high frequency because of lead inductance, so open-circuit z measurements are preferred there and short-circuit y measurements at low frequency. Third, algebraic convenience: ABCD matrices multiply under cascading while z and y do not, so a chain of stages is trivial in ABCD and laborious otherwise.

Point required	Marks
Existence argument with a concrete example	1
Measurement argument	1
Operation-convenience argument	1
Total	3

B5 (2 marks) — CO5, L4, Ch.17, pp.716-720

Question. State the ABCD defining equations, explain the minus signs, and give the matrix for a series impedance Z .

Model answer. $V_1 = A \cdot V_2 - B \cdot I_2$ and $I_1 = C \cdot V_2 - D \cdot I_2$. The minus signs appear because the standard convention takes I_2 as entering the network while the transmission description treats the current as

flowing out toward the load. For a lone series impedance, opening port 2 gives $V_1 = V_2$ so $A = 1$ and $I_1 = 0$ so $C = 0$; shorting port 2 gives $V_1 = -I_2 Z$ so $B = Z$, and $I_1 = -I_2$ so $D = 1$. The matrix is $\begin{bmatrix} 1, Z \\ 0, 1 \end{bmatrix}$, and $AD - BC = 1$ as required.

Point required	Marks
Correct equations with the sign explanation	1
Correct matrix with its derivation	1
Total	2

B6 (2 marks) — CO5, L4, Ch.17, pp.716-720

Question. Cascade a series $10\ \Omega$ with a shunt $0.05\ S$ and verify the result.

Model answer. The series element is $\begin{bmatrix} 1, 10 \\ 0, 1 \end{bmatrix}$ and the shunt is $\begin{bmatrix} 1, 0 \\ 0.05, 1 \end{bmatrix}$. Multiplying in signal-flow order gives $\begin{bmatrix} 1 + 10(0.05), 10 \\ 0.05, 1 \end{bmatrix} = \begin{bmatrix} 1.5, 10 \\ 0.05, 1 \end{bmatrix}$. The check $AD - BC = 1.5 \times 1 - 10 \times 0.05 = 1$ confirms reciprocity. Note that the product already accounts for the loading of the shunt on the series element, which naive transfer-function multiplication would miss.

Point required	Marks
Correct individual matrices and ordered product	1
Correct reciprocity check with the loading remark	1
Total	2

B7 (2 marks) — CO5, L4, Ch.17, pp.699-720

Question. Distinguish reciprocity from symmetry, giving the condition for each in z and in ABCD parameters.

Model answer. Reciprocity means source and meter may be interchanged with the same reading; in z parameters it is $z_{12} = z_{21}$ and in ABCD it is $AD - BC = 1$. Symmetry means the two ports are interchangeable — the network looks identical from either side; in z it is $z_{11} = z_{22}$ and in ABCD it is $A = D$. Every symmetric passive network is reciprocal, but an unbalanced ladder is reciprocal without being symmetric, so the converse does not hold.

Point required	Marks
Both conditions stated correctly in both parameter sets	1
Correct statement of the one-way implication with an example	1
Total	2

B8 (2 marks) — CO5, L4, Ch.17, pp.699-708

Question. Derive the π equivalent from $y = \begin{bmatrix} 0.15, -0.1 \\ -0.1, 0.15 \end{bmatrix} S$.

Model answer. The series admittance is $-y_{12} = 0.1\ S$, which is $10\ \Omega$. The port 1 shunt admittance is $y_{11} + y_{12} = 0.15 - 0.1 = 0.05\ S$, which is $20\ \Omega$, and the port 2 shunt is $y_{22} + y_{12} = 0.05\ S$, also $20\ \Omega$. Since $y_{12} = y_{21}$ the network is reciprocal, so three passive branches suffice. The π model is therefore $20\ \Omega$ shunt, $10\ \Omega$ series, $20\ \Omega$ shunt.

Point required	Marks
Correct series admittance from $-y_{12}$	1
Correct shunt values with the reciprocity check	1
Total	2

B9 (2 marks) — CO5, L3, Ch.17, pp.713-716

Question. Explain the h parameters and why their mixed units are unavoidable.

Model answer. They relate mixed variables: $V_1 = h_{11}I_1 + h_{12}V_2$ and $I_2 = h_{21}I_1 + h_{22}V_2$. Because a voltage is expressed in terms of a current, h_{11} must be in ohms; because a current is expressed in terms of a voltage, h_{22} must be in siemens; h_{12} and h_{21} are ratios of like quantities and are dimensionless. The mixed choice is not arbitrary — it matches how a bipolar transistor is used and measured, which is why datasheets quote h_{ie} and h_{fe} directly.

Point required	Marks
Correct equations with the units of all four	1
Correct explanation linking the variable choice to transistor use	1
Total	2

B10 (2 marks) — CO5, L4, Ch.17, pp.708-713

Question. Compute Z_{in} for $z = [[30, 20], [20, 30]] \Omega$ with a 50Ω load, and verify with both limiting cases.

Model answer. $Z_{in} = z_{11} - z_{12}z_{21}/(z_{22} + Z_L) = 30 - (20 \times 20)/(30 + 50) = 30 - 5 = 25 \Omega$. As $Z_L \rightarrow \infty$ the reflected term vanishes and $Z_{in} \rightarrow z_{11} = 30 \Omega$, which is the open-circuit definition. As $Z_L \rightarrow 0$ the reflected term becomes $400/30 = 13.3 \Omega$, giving $Z_{in} = 16.7 \Omega$, which matches a direct short-circuit calculation on the T network. So Z_{in} ranges from 16.7Ω to 30Ω purely with the load.

Point required	Marks
Correct Z_{in} from the formula	1
Both limiting cases verified	1
Total	2

B11 (2 marks) — CO5, L5, Ch.17, pp.708-713

Question. Compute Z_{out} for $z = [[30, 20], [20, 30]] \Omega$ driven from a 10Ω source, and state the matched load.

Model answer. $Z_{out} = z_{22} - z_{12}z_{21}/(z_{11} + Z_S) = 30 - 400/(30 + 10) = 30 - 10 = 20 \Omega$. For maximum power transfer into a resistive load, the load should equal 20Ω ; for a complex Z_{out} the load would be its conjugate. Checking the limit $Z_S \rightarrow \infty$, the reflected term vanishes and $Z_{out} \rightarrow z_{22} = 30 \Omega$, which is the open-circuit output impedance.

Point required	Marks
Correct Z_{out}	1
Correct matched load with the limiting-case check	1

Point required	Marks
Total	2

B12 (2 marks) — CO5, L5, Ch.17, pp.692-713

Question. Explain why a short circuit becomes unusable as a measurement termination at high frequency, with a numerical estimate.

Model answer. A physical short has lead inductance, so its impedance is $j\omega L$ rather than zero and rises with frequency. With a typical 5 nH lead at 100 MHz, $Z = j \cdot 2\pi \cdot 10^8 \cdot 5 \times 10^{-9} = j3.14 \Omega$. Against a 50Ω reference that is a perturbation of about 6 %, and it grows linearly with frequency. Since y parameters require a true short, they become progressively unreliable, which is why scattering parameters using matched terminations are used instead at these frequencies.

Point required	Marks
Correct identification of lead inductance as the cause	1
Correct numerical estimate and the consequence for parameter choice	1
Total	2

B13 (2 marks) — CO5, L5, Ch.17, pp.708-720

Question. For $z = [[30, 20], [20, 30]] \Omega$ with $Z_S = 10 \Omega$ and $Z_L = 50 \Omega$, compute the voltage gain and comment on the output match.

Model answer. $A_v = z_{21}Z_L / (z_{11}(z_{22} + Z_L) - z_{12}z_{21}) = 20 \times 50 / (30 \times 80 - 400) = 1000 / 2000 = 0.5$. The network attenuates, as expected for a passive T. The output impedance with this source is $Z_{out} = 30 - 400 / 40 = 20 \Omega$, so the 50Ω load is mismatched and delivers less than the maximum available output power; a 20Ω load would be matched. This is the Module 2 maximum power transfer result expressed in two-port terms.

Point required	Marks
Correct voltage gain	1
Correct output-match comment linked to Z_{out}	1
Total	2